

Universal small-spin formulas for the shadow area, the capture cross-section and the spin-down of Kerr-like rotating black holes

Jacob Goodchild*

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Abstract

Many rotating black-hole metrics studied in the literature are generated from a static, spherically symmetric seed $ds^2 = -f dt^2 + dr^2/f + r^2 d\Omega^2$ by the Newman–Janis algorithm in the form of Azreg-Aïnou, which yields the Kerr metric with $2Mr$ replaced by $r^2(1 - f(r))$. For this whole class we give closed formulas, in terms of f and its derivatives at a single radius, for the leading effects of a small spin a on three observables: (i) the area of the black-hole shadow seen by a distant observer at any inclination, to order a^2 and, with a term-by-term finite-part method, to orders a^4 , a^6 and a^8 ; (ii) the capture cross-section for neutral massive particles of any speed arriving from any direction, to order a^2 (and to order a^4 along the spin axis); and (iii) the mean angular momentum swallowed per unit captured energy, which controls the spin-down of a black hole accreting collisionless matter. The order- a^2 shadow formula depends on f only through $f(R)$ and the curvature combination $\kappa = 2f - R^2 f''$ at the photon sphere R ; the massive-particle formula contains the combination $3ff' + rff'' - 2rf'^2$ whose zero defines the innermost stable circular orbit. Specialising to Kerr–Newman gives explicit expressions valid at arbitrary charge, including the shadow area to order a^4 ; for Kerr the method reproduces the known low-order terms and yields the order- a^6 and a^8 coefficients of the shadow area at any inclination. All results were checked against direct high-precision computations of the exact shadow contour or of the exact capture boundary for Kerr, Kerr–Newman, rotating Bardeen and rotating Hayward black holes, with agreement to 11–22 significant digits.

1 Introduction

The shadow of a black hole—the dark region in the sky bounded by the image of the unstable photon orbits—is now an observable [1]. For the Kerr metric the boundary of the shadow is known in closed parametric form [2, 3]. A large body of work constructs rotating counterparts of non-Kerr static black holes (regular black holes, black holes surrounded by matter, quantum-corrected black holes, and so on) with the Newman–Janis algorithm [4], most often in the non-complexified form of Azreg-Aïnou [5], and then computes their shadows. For this class the geodesic equations separate and general formulas for the shadow contour are available [6, 7]; observables such as the area, the circumference or the displacement of the shadow are then usually computed numerically, model by model.

The purpose of this paper is to give *universal* closed formulas for the small-spin behaviour of such observables, valid for any seed function $f(r)$. The main results are:

- the order- a^2 correction to the shadow area at any inclination, eq. (4.1);
- the order- a^4 correction, eqs. (4.2)–(4.5), obtained with a finite-part (residue at infinity) method that extends mechanically to a^6 and a^8 ;

*Independent researcher. Contact: jacob.l.goodchild@gmail.com. This work was carried out with substantial assistance from an AI system; see the AI disclosure statement at the end of the paper.

- the order- a^2 correction to the capture cross-section of neutral massive particles of arbitrary speed and direction, eq. (4.6), and its on-axis continuation to order a^4 , eq. (4.7);
- the mean angular momentum per unit energy carried into the hole by captured particles, eq. (4.8).

For Kerr and Kerr–Newman these specialise to explicit rational functions of the radius of the relevant circular orbit (Section 5). Several of these special cases were known before, in particular for Kerr [9, 10, 12, 15]; Section 7 lists precisely which results are previously known and which, to the best of our knowledge, are new.

Units are $G = c = 1$. Lengths are measured in units of the mass parameter of the seed metric where relevant; all formulas below are invariant under the rescaling $r \rightarrow \lambda r$, $a \rightarrow \lambda a$, $\text{area} \rightarrow \lambda^2 \text{area}$, with $f(r)$ rescaled accordingly.

2 Setting

2.1 Metric and geodesics

Let $f(r)$ be a smooth function and put

$$\Sigma = r^2 + a^2 \cos^2 \vartheta, \quad \Delta = r^2 f(r) + a^2. \quad (2.1)$$

The rotating metrics considered here are

$$ds^2 = -\left(1 - \frac{r^2(1-f)}{\Sigma}\right)dt^2 - \frac{2ar^2(1-f)\sin^2\vartheta}{\Sigma}dt d\varphi + \frac{\Sigma}{\Delta}dr^2 + \Sigma d\vartheta^2 + \frac{[(r^2 + a^2)^2 - \Delta a^2 \sin^2\vartheta]\sin^2\vartheta}{\Sigma}d\varphi^2, \quad (2.2)$$

which for $f = 1 - 2M/r$ is the Kerr metric and for $f = 1 - 2M/r + Q^2/r^2$ the Kerr–Newman metric. This is the form produced by the Newman–Janis algorithm without complexification [5] when the seed is $ds^2 = -f dt^2 + dr^2/f + r^2 d\Omega^2$. The Hamilton–Jacobi equation separates, and for a geodesic with energy E , axial angular momentum L , Carter constant Q and mass μ (all per unit mass for $\mu = 1$; $\mu = 0$ for light),

$$\Sigma \frac{dr}{d\lambda} = \pm \sqrt{\mathcal{R}(r)}, \quad \mathcal{R}(r) = [E(r^2 + a^2) - aL]^2 - \Delta[\mu^2 r^2 + (L - aE)^2 + Q], \quad (2.3)$$

$$\Sigma \frac{d\vartheta}{d\lambda} = \pm \sqrt{\Theta(\vartheta)}, \quad \Theta(\vartheta) = Q - \cos^2 \vartheta \left[a^2(\mu^2 - E^2) + \frac{L^2}{\sin^2 \vartheta} \right]. \quad (2.4)$$

2.2 The impact plane

Consider particles that arrive from infinity (or reach a distant observer) along the direction making angle ϑ_o with the spin axis, and write

$$C \equiv \cos^2 \vartheta_o. \quad (2.5)$$

With $p = \sqrt{E^2 - \mu^2}$ the asymptotic momentum, the Cartesian impact parameters (celestial coordinates for light) are $\alpha = -L/(p \sin \vartheta_o)$, $\beta = \pm \sqrt{\Theta(\vartheta_o)}/p$, so that

$$b^2 \equiv \alpha^2 + \beta^2 = \frac{L^2 + Q}{p^2} + a^2 C. \quad (2.6)$$

The *shadow* (for light) and the *capture region* (for massive particles of given E) are the regions of the (α, β) plane whose geodesics reach the horizon. Their boundaries are traced by orbits on which $\mathcal{R}$ has a double root, $\mathcal{R} = \mathcal{R}' = 0$; we write $\mathcal{A}$ for the area of the shadow and σ for the capture cross-section.

2.3 The non-rotating data

At $a = 0$ the boundary is a circle. For light it is set by the photon sphere $r = R$,

$$Rf'(R) = 2f(R), \quad \mathcal{A}_0 = \frac{\pi R^2}{f(R)}. \quad (2.7)$$

For massive particles with specific energy $E > 1$ (speed v at infinity, $E^2 = 1/(1 - v^2)$) it is set by the unstable circular orbit of that energy, of radius r , with

$$E^2 = \frac{2f^2}{2f - rf'}, \quad L^2 = \frac{r^3 f'}{2f - rf'}, \quad \sigma_0 = \frac{\pi L^2}{E^2 - 1} = \frac{\pi r^3 f'}{2f^2 - 2f + rf'}, \quad (2.8)$$

where f and its derivatives are evaluated at r . These relations are standard; for the Schwarzschild metric they give the classical capture cross-section [18] $\sigma_0 = \pi r^3/(4 - r)$ (with $M = 1$), which tends to $16\pi/v^2$ for slow particles and to 27π for light. Throughout, r denotes this circular-orbit radius, which lies between the photon sphere (light) and the marginally bound orbit ($v \rightarrow 0$).

We use the abbreviations

$$\kappa \equiv 2f - R^2 f''(R) \quad (\text{at the photon sphere}), \quad J \equiv 3ff' + rf f'' - 2rf'^2 \quad (\text{at radius } r). \quad (2.9)$$

For static metrics, κ is related to the Lyapunov exponent λ of the photon sphere (in coordinate time) by $\lambda^2 = f\kappa/(2R^2)$ [8], and $J = 0$ is the condition for the innermost stable circular orbit. For Schwarzschild, $\kappa = 2$ and $J = 2(r - 6)/r^3$.

3 Method

3.1 Perturbation of the double-root conditions

For the order- a^2 results we parametrise the boundary orbits by their total angular momentum $\ell = \sqrt{L^2 + Q}$ and the inclination $\cos i = L/\ell$, and expand the critical values

$$\ell_c = \ell_0 + a\ell_1 \cos i + a^2 \ell_2 (\cos^2 i) + O(a^3), \quad r_c = r + O(a), \quad (3.1)$$

by solving $\mathcal{R} = \mathcal{R}' = 0$ order by order, with f replaced by its Taylor series about the unperturbed radius. To leading order the inclination is related to the polar angle ϕ in the impact plane by $\cos i = -\sin \vartheta_o \cos \phi$, with ϕ measured from the α axis. The area of a star-shaped region is $\frac{1}{2} \oint b^2 d\phi$, and by (2.6), $b^2 = \ell_c^2/p^2 + a^2 C$. Averaging over ϕ gives the order- a^2 coefficient; the $O(a^2)$ reparametrisation between ϕ and i does not contribute at this order because the $O(a)$ term is odd in $\cos \phi$. In particular one finds the universal first-order coefficient

$$\ell_1 = -E \frac{1 - f}{f} \quad (3.2)$$

(sign chosen so that prograde orbits have $\cos i > 0$). The derivative f''' drops out of all order- a^2 results. The computations were done with SYMPY.

3.2 Finite-part expansion of the shadow area

For light the boundary of the shadow is known in closed parametric form for any $\Delta(r)$ [6]:

$$\xi(r) = \frac{(r^2 + a^2)\Delta' - 4r\Delta}{a\Delta'}, \quad \eta(r) = \frac{r^2[16a^2\Delta - (4\Delta - r\Delta')^2]}{a^2\Delta'^2}, \quad (3.3)$$

with $\xi = L/E$, $\eta = Q/E^2$, $\alpha = -\xi/\sin \vartheta_o$ and $\beta^2 = \eta + a^2 C - \xi^2 \cot^2 \vartheta_o$. Set $r = R + ax$. Because $4\Delta - r\Delta' = r^2(2f - rf') + 4a^2$ vanishes at $a = 0$, $r = R$, every coefficient of the expansion

of ξ and η in powers of a at fixed x is a *polynomial* in x . After the linear change of variable $u = \alpha_0(x)$ (the zeroth-order abscissa), $\beta^2 = b_0^2 - u^2 + \delta(u, a)$ with $b_0^2 = R^2/f$ and δ a polynomial in u at every order in a . The area $\mathcal{A} = 2 \int \beta d\alpha$ is an analytic function of a , and its expansion can be computed term by term with the finite-part (Hadamard) rule

$$\text{FP} \int_{-b_0}^{b_0} u^m (b_0^2 - u^2)^{\frac{1}{2}-j} du = -\pi (-1)^j \binom{\frac{1}{2}-j}{n} (-b_0^2)^n, \quad n = \frac{m+2-2j}{2}, \quad (3.4)$$

for even m with $n \geq 0$ (and zero otherwise), which follows from the residue of $u^m (b_0^2 - u^2)^{1/2-j}$ at infinity. Expanding $\sqrt{b_0^2 - u^2 + \delta} d\alpha/du$ binomially in δ and applying (3.4) to each monomial gives the area to any order. With truncated power series over the rational function field in $f(R), Rf'(R), \dots$ this takes about one second at order a^4 , ten seconds at order a^6 and seven minutes at order a^8 on a laptop-class CPU.

4 Results

Throughout this section $f, f', f'', \dots$ denote values at the photon sphere R (for the shadow) or at the circular-orbit radius r (for massive particles), and $C = \cos^2 \vartheta_o$.

4.1 Shadow area to order a^2

Result 1 (Shadow area).

$$\mathcal{A} = \frac{\pi R^2}{f} \left[1 + a^2 \left\{ (1-C) \left(\frac{1}{2fR^2} - \frac{4}{R^2\kappa} \right) + C \frac{2f-1}{fR^2} \right\} + O(a^4) \right]. \quad (4.1)$$

Corollary 1 (Polar observer). *For $\vartheta_o = 0$ the shadow is a disc, of area $\mathcal{A} = \pi[R^2/f + a^2(2f-1)/f^2] + O(a^4)$. (The exact polar radius of integrable spinning spacetimes is given in [14]; this is its small-spin expansion.)*

Corollary 2 (Displacement and circumference). *To first order in a the shadow is displaced along the α axis by $a \sin \vartheta_o (1-f)/f$ (for Kerr: $2a \sin \vartheta_o$, a known result [9, 10]). To order a^2 the boundary is a displaced circle plus a $\cos 2\phi$ deformation, which changes neither area nor length at this order; hence the circumference is $P = 2\pi R f^{-1/2} [1 + \frac{1}{2}a^2\{\dots\} + O(a^4)]$ with the same bracket $\{\dots\}$ as in (4.1).*

4.2 Shadow area to order a^4

Write $f_3 = R^3 f'''(R)$ and $f_4 = R^4 f''''(R)$.

Result 2 (Next order).

$$\mathcal{A} = \frac{\pi R^2}{f} \left[1 + a^2 K_2 + \frac{a^4}{R^4} \frac{N_0 + N_1 C + N_2 C^2}{8f^2 \kappa^5} + O(a^6) \right], \quad (4.2)$$

where K_2 is the bracket of (4.1) and

$$N_0 = -\kappa^5 + 16f\kappa^4 - 160f^2\kappa^3 + 32f^2f_3\kappa^2 - 384f^3\kappa^2 + 256f^3f_3\kappa - 32f^3f_4\kappa - 96f^3f_3^2, \quad (4.3)$$

$$N_1 = 2[-3\kappa^5 + 4f\kappa^5 + 32f\kappa^4 - 32f^2\kappa^4 - 160f^2\kappa^3 + 32f^2f_3\kappa^2 + 192f^3\kappa^3 + 384f^3\kappa^2 - 64f^3f_3\kappa^2 - 256f^3f_3\kappa + 32f^3f_4\kappa + 96f^3f_3^2], \quad (4.4)$$

$$N_2 = 15\kappa^5 - 24f\kappa^5 - 144f\kappa^4 + 8f^2\kappa^5 + 192f^2\kappa^4 + 480f^2\kappa^3 - 96f^2f_3\kappa^2 - 64f^3\kappa^4 - 384f^3\kappa^3 - 384f^3\kappa^2 + 128f^3f_3\kappa^2 + 256f^3f_3\kappa - 32f^3f_4\kappa - 96f^3f_3^2. \quad (4.5)$$

The coefficients of a^6 and a^8 are obtained by the same procedure; they are long rational expressions in $f, R^k f^{(k)}(R)$ ($k \leq 6$ and $k \leq 8$) and C , and are provided in machine-readable form with the accompanying code (see Code and data availability).

4.3 Capture cross-section to order a^2 , any speed

Result 3 (Capture). *For neutral particles with speed v at infinity, arriving at angle ϑ_o to the spin axis, with r the radius of the unstable circular orbit of energy $E = (1 - v^2)^{-1/2}$ in the seed metric [eq. (2.8)],*

$$\sigma = \sigma_0 \left[1 + a^2 \left\{ (1 - C) \left(\frac{1}{2fr^2} + \frac{f'^2}{rfJ} \right) + C \frac{2f + rf' - 2}{r^3 f'} \right\} + O(a^4) \right]. \quad (4.6)$$

At the photon sphere ($rf' = 2f$, hence $J = -f\kappa/r$) the bracket reduces to that of (4.1), as it must.

Result 4 (On the spin axis to order a^4). *For $\vartheta_o = 0$ the capture region is exactly a disc of radius² $Q/p^2 + a^2$ (with $L = 0$), and*

$$\sigma = \sigma_0 \left[1 + a^2 \frac{2f + rf' - 2}{r^3 f'} + a^4 \frac{2(1 - f)^2(3f' + rf'')}{r^5 f' J} + O(a^6) \right]. \quad (4.7)$$

4.4 Angular momentum swallowed

Let $\langle L \rangle$ be the mean axial angular momentum of the captured particles (averaged with uniform weight over the capture region of the impact plane) and E their energy, both per unit rest mass. Then $\langle L \rangle / E = dJ_{\text{BH}} / dM_{\text{BH}}$ is the angular momentum gained per unit mass-energy gained.

Result 5 (Spin-down).

$$\frac{\langle L \rangle}{E} = -a \sin^2 \vartheta_o \frac{1 - f}{f} + O(a^3), \quad (4.8)$$

for massive particles of any speed (f at r) and for light (f at R).

For a black hole immersed in an isotropic flux of particles of a single speed, averaging over directions with the leading-order weight σ_0 gives $\langle L \rangle / E = -\frac{2}{3}a(1 - f)/f$ and, for the dimensionless spin $a_* = J_{\text{BH}} / M_{\text{BH}}^2$,

$$\frac{da_*}{d \ln M_{\text{BH}}} = -2a_* - \frac{2a_*}{3} \frac{1 - f}{f} + O(a_*^3). \quad (4.9)$$

5 Special cases

5.1 Kerr

With $f = 1 - 2/r$ ($M = 1$): $R = 3$, $f(R) = 1/3$, $\kappa = 2$, $f_3 = 4$, $f_4 = -16$. The particle speed is encoded by the circular-orbit radius

$$r = \frac{4v^2 - 1 + \sqrt{1 + 8v^2}}{2v^2} \in [3, 4], \quad \sigma_0 = \frac{\pi r^3}{4 - r}. \quad (5.1)$$

Results 1–5 give the following. The a^2 and a^4 terms of (5.2) are known [9, 10, 12]; the a^6 and a^8 terms do not appear in the literature we found. The capture formula (5.3) at order a^2 and the order- a term of $\langle L \rangle / E$ follow from the slow-rotation expansion of the critical Carter constant of Mach, Momennia and Sarbach [15] (who display the direction-averaged versions), and agree

with the fitted coefficients of [17]:

$$\frac{\mathcal{A}}{27\pi} = 1 - \frac{(1+C)a^2}{18} - \left(\frac{1}{72} + \frac{5C}{972} - \frac{5C^2}{1944}\right)a^4 - \left(\frac{2059}{314928} + \frac{167C}{104976} - \frac{185C^2}{104976} - \frac{5C^3}{3888}\right)a^6 - \left(\frac{88175}{22674816} + \frac{3869C}{5668704} - \frac{4865C^2}{3779136} - \frac{83C^3}{69984} - \frac{667C^4}{7558272}\right)a^8 + O(a^{10}), \quad (5.2)$$

$$\frac{\sigma}{\sigma_0} = 1 - a^2 \frac{r+3C(4-r)}{2r^2(6-r)} + O(a^4), \quad \frac{\sigma(\vartheta_o=0)}{\sigma_0} = 1 - \frac{a^2}{r^2} - \frac{4a^4}{r^4(6-r)} + O(a^6), \quad (5.3)$$

$$\frac{\langle L \rangle}{E} = -\frac{2a \sin^2 \vartheta_o}{r-2} + O(a^3). \quad (5.4)$$

For light ($r=3$) the capture result reproduces the a^2 term of (5.2); for slow particles ($r \rightarrow 4$) the a^2 correction $-a^2/16$ is independent of direction.

Higher orders for Kerr. By fitting high-precision evaluations of the exact Kerr capture region at radii $r \in \{15/4, 7/2, 10/3, 13/4, 18/5, 16/5\}$ we identified exact rational coefficients for the full order- a^4 term and for the order- a^3 term of the swallowed angular momentum:

$$\frac{\sigma}{\sigma_0} = 1 - a^2 \frac{r+3C(4-r)}{2r^2(6-r)} + a^4 \frac{\alpha_8 + \beta_8 C + \gamma_8 C^2}{8r^4(6-r)^5} + O(a^6), \quad (5.5)$$

$$\alpha_8 = -r^2(2r^3 + 42r^2 - 549r + 1458), \quad \beta_8 = 6r(2r^4 + 2r^3 - 239r^2 + 1302r - 2016),$$

$$\gamma_8 = -10r^5 - 2r^4 + 1653r^3 - 13266r^2 + 39744r - 41472,$$

$$\frac{\langle L \rangle}{E} = -\frac{2a(1-C)}{r-2} + a^3(1-C) \frac{\frac{3}{2}(r-5) - C \frac{7r^2-63r+144}{2r}}{(r-2)(6-r)^3} + O(a^5). \quad (5.6)$$

Eq. (5.5) satisfies $\alpha_8 + \beta_8 + \gamma_8 = -32(6-r)^4$, reproducing the on-axis result (5.3), which was derived analytically; at $r=3$ it reproduces the a^4 term of (5.2). Eq. (5.6) was found from data at $C=0$ and $C=1/2$ and then confirmed at $C=1/4$ and $C=4/5$; both formulas were further confirmed at radii and angles not used in their identification ($r=3.55$, 40° and $r=3.15$, 70° ; Table 1). These two formulas are identifications supported by strong numerical evidence, not derivations. The order- a^4 capture term is beyond the expansion of [15], which is given to third order in the spin, and has only been available as numerical fits [17]; the a^3 term of $\langle L \rangle/E$ can in principle be derived from the third-order terms of [15], where it is not displayed. Averaging (5.6) over an isotropic flux, with the order- a^2 weight (5.3), gives

$$\left. \frac{\langle L \rangle}{E} \right|_{\text{iso}} = -\frac{4a}{3(r-2)} + \frac{4a^3(3r^3 - 19r^2 + 48r - 144)}{15r^2(r-2)(6-r)^3} + O(a^5), \quad (5.7)$$

equal to $-(2a/3 + a^3/15)$ for slow particles, as found in [15, 16], and $-(4a/3 + 8a^3/81)$ for light.

5.2 Kerr–Newman

With $f = 1 - 2/r + q^2/r^2$ ($M=1$, charge q ; neutral test particles):

$$\frac{\sigma}{\sigma_0} = 1 - a^2 \frac{r^2(r-q^2) - C(3r^3 - 12r^2 - q^2r^2 + 18q^2r - 8q^4)}{2r(r-q^2)(6r^2 - r^3 - 9q^2r + 4q^4)} + O(a^4), \quad (5.8)$$

$$\sigma_0 = \frac{\pi r^4(r-q^2)}{4r^2 - r^3 - 4q^2r + q^4},$$

$$\frac{\langle L \rangle}{E} = -a \sin^2 \vartheta_o \frac{2r-q^2}{r^2-2r+q^2} + O(a^3), \quad (5.9)$$

with $E^2 = (r^2 - 2r + q^2)^2 / [r^2(r^2 - 3r + 2q^2)]$. The factor $r^3 - 6r^2 + 9q^2r - 4q^4$ is the Reissner–Nordström ISCO polynomial. For the shadow, with $R = \frac{1}{2}(3 + \sqrt{9 - 8q^2})$,

$$\mathcal{A} = \frac{\pi R^4}{R^2 - 2R + q^2} \left[1 - a^2 \frac{R + 3C(R - 2)}{R^2(R - 1)(2R - 3)} + a^4 K_4 + O(a^6) \right], \quad (5.10)$$

where

$$K_4 = \frac{1}{2R^4(R - 1)^2(2R - 3)^5} \left[C^2(R^6 - 38R^5 + 286R^4 - 768R^3 + 774R^2 - 108R - 162) \right. \\ \left. - C(18R^6 - 108R^5 + 288R^4 - 492R^3 + 504R^2 - 216R) \right. \\ \left. - 15R^6 + 58R^5 - 78R^4 + 36R^3 \right].$$

Expanding in q reproduces the double series $\mathcal{A}/(27\pi) = 1 - q^2/3 - (1 + C)a^2/18 - q^4/27 - (3 + C)a^2q^2/81 + \dots$, which is contained in the perturbative results of Kobialko and Gal'tsov [12]; eqs. (5.8)–(5.10) are exact in q .

5.3 Regular black holes

For the rotating Bardeen ($f = 1 - 2r^2/(r^2 + g^2)^{3/2}$ [20]) and Hayward ($f = 1 - 2r^2/(r^3 + 2l^2)$ [21]) black holes, the order- a^2 coefficient of $\mathcal{A}/\mathcal{A}_0$ at $\vartheta_o = 90^\circ$ is -0.0797470530672 ($g = 0.4$) and -0.0863835115987 ($l = 0.5$) respectively; further values are listed in Table 1.

6 Verification

Every result was checked against an independent computation that does not use the perturbative expansions.

- *Shadow*: the exact contour (3.3) was integrated by tanh–sinh quadrature at 40 digits for $a = 0.01, \dots, 0.05$, and the coefficients of a^2 and a^4 were extracted by polynomial fitting in a^2 .
- *Capture*: the exact boundary $b(\phi)$ was computed at 48 impact-plane angles by solving $\mathcal{R} = \mathcal{R}' = 0$ with Newton's method at 40 digits, the area evaluated as $\frac{1}{2} \oint b^2 d\phi$ (trapezoidal rule, spectrally accurate for periodic integrands), and the coefficients extracted as above. The swallowed angular momentum was computed as the first moment of the same region.
- *Symbolic*: the Kerr–Newman specialisation of (4.6) was checked to agree identically with an independent Kerr–Newman calculation; the light limit of (4.7) agrees with (4.2) at $C = 1$ (exactly for Kerr, where both give $-4a^4/243$, and to 10^{-25} for Bardeen); the Kerr limits of the a^6 and a^8 shadow coefficients agree exactly with rational coefficients identified independently from 250-digit evaluations of an exact elliptic-integral formula for the edge-on Kerr shadow area and from a single-integral representation at general inclination.

7 Relation to previous work

To the best of our knowledge, based on searches of arXiv, INSPIRE and the wider web (October 2026), the status of the results is as follows.

Previously known. (i) The exact shadow contour for metrics of the form (2.2) [6, 7], and the exact polar shadow radius for general integrable spinning spacetimes [14]. (ii) For Kerr, the order- a^2 shape of the shadow (an ellipse displaced by $2a \sin \vartheta_o$) [9, 10, 11], hence the a^2 term of (5.2); the edge-on area to order a^2 is stated explicitly in [13]. (iii) Kobialko and

Quantity	Case	Direct computation	Formula
a^2 coeff. of $\mathcal{A}/\mathcal{A}_0$	Bardeen $g = 0.4$, 90°	−0.07974705306715083	−0.07974705306715083
	Bardeen $g = 0.4$, 50°	−0.1035696370499696	−0.1035696370499696
	Hayward $l = 0.5$, 90°	−0.08638351159869441	−0.08638351159869439
	Hayward $l = 0.5$, 50°	−0.10712747758	−0.10712747758
	Kerr, 50°	−0.07850977284258527	−0.07850977284258527
a^4 coeff. of $\mathcal{A}/\mathcal{A}_0$	Bardeen $g = 0.4$, 90°	−0.0317475715025	−0.0317475715025
	Bardeen $g = 0.4$, 50°	−0.0319437641728	−0.0319437641729
	Kerr–Newman $q = 0.6$, 60°	−0.0344681711527	−0.0344681711528
	Kerr–Newman $q = 0.8$, 90°	−0.0918348939951	−0.0918348939966
a^2 coeff. of σ/σ_0	Bardeen, $r = 3.4$, 60°	−0.08155531033444	−0.08155531033444
	Bardeen, $r = 3.1$, 90°	−0.07771672446027	−0.07771672446026
	Hayward, $r = 3.4$, 60°	−0.08231846186746	−0.08231846186746
	Hayward, $r = 3.1$, 90°	−0.08168373732763	−0.08168373732763
	KN $q = 0.5$, $r = 3.5$, 60°	−0.0790432393693279	−0.0790432393693263
	KN $q = 0.8$, $r = 3.0$, 45°	−0.136651895747247	−0.136651895747233
	Bardeen, $r = 3.4$	−0.0938935339679, −0.0161005095929	same to all digits
on-axis a^2 , a^4 coeff.	Bardeen, $r = 3.1$	−0.114811334175, −0.0214112232194	same to all digits
	Bardeen, $r = 3.4$, 60°	−1.01983100502	−1.01983100502
$\langle L \rangle / (aE)$	KN $q = 0.5$, $r = 3.5$, 60°	−0.920454545455	−0.920454545455
a^3 coeff. of $\langle L \rangle / E$, Kerr	$r = 7/2$, $C = 1/4$	−0.0825714285714285714	−0.0825714285714285714
	$r = 10/3$, $C = 4/5$	−0.030955078125	−0.030955078125
	$r = 3.55$, 40°	−0.05225970864164288	−0.05225970864164288
	$r = 3.55$, 40°	−0.01290083149371615	−0.01290083149371615
a^4 coeff. of σ/σ_0 , Kerr, eq. (5.5)	$r = 3.15$, 70°	−0.01428654979243309	−0.01428654979243309

Table 1: Selected comparisons between the formulas and direct high-precision computations (values from `verify_paperA.log`, except the Hayward 50° shadow row and the Bardeen $r = 3.1$ and Hayward $r = 3.4$ capture rows, which come from the earlier scripts `univ_check_light.py` and `univ_check_mass.py`). Bardeen: $g = 0.4$; Hayward: $l = 0.5$ ($M = 1$). The direct computations use the exact contour or the exact capture boundary, not the expansions.

Gal’tsov [12] developed a perturbation theory for shadows of Kerr-like spacetimes, expanding around Schwarzschild to fifth order in the spin and to second order in a small parameter measuring the deviation from Kerr. Their results, through their areal definition of the mean radius, yield the order- a^4 Kerr area of (5.2) and the Kerr–Newman double series to combined fourth order. (iv) For Kerr, the order- a^2 capture cross-section at any speed and direction, and the swallowed angular momentum, follow from the slow-rotation expansion of the critical Carter constant to $O(a^3)$ by Mach, Momennia and Sarbach [15], who give the direction-averaged accretion rates explicitly (see also [16]); Karydas, Vicente and Bertone computed the direction- and speed-dependent coefficients through order a^4 as numerical fits [17]. The Schwarzschild cross-section $\sigma_0(v)$ is classical [18], and the Reissner–Nordström one at arbitrary speed was studied in [19].

New in this paper, as far as we could determine. (a) The formulas exact in f : Results 1–5, including the order- a^4 shadow formula, the a^6 and a^8 coefficients, the capture cross-section of massive particles for general f , its on-axis continuation to a^4 and the angular-momentum law $-a \sin^2 \vartheta_o (1 - f)/f$. (b) For Kerr: the a^6 and a^8 shadow-area coefficients at general inclination, the full order- a^4 capture term (5.5) (numerically identified), and the explicit direction-dependent order- a^3 swallowed angular momentum (5.6) (numerically identified, and derivable in principle from [15]). (c) For Kerr–Newman: the closed forms exact in the charge, eqs. (5.8)–(5.10).

8 Discussion

The formulas above reduce the small-spin behaviour of the shadow area, the capture cross-section and the accreted angular momentum of any Kerr-like black hole of the form (2.2) to the evaluation of f and a few derivatives at one radius. Two limitations should be kept in mind. First, the metrics (2.2) are generated by an algorithm and are in general not solutions of any particular set of field equations; the results apply to them as geometries. Rotating metrics of

other forms (for example the Kerr–Sen metric, whose Σ differs) are not covered. Second, the results are expansions in a ; their accuracy for rapidly rotating holes should be judged from the size of the successive terms, which we provide up to a^8 for the shadow.

AI disclosure statement

This work was carried out by the author with substantial assistance from an AI system, Claude (Anthropic), used through the Claude Code software. Under the author’s direction, the AI system proposed the research directions, wrote and ran the computer-algebra and numerical code (Python with SymPy and mpmath), performed the derivations and checks reported here, searched the literature, and drafted the text of this paper. The author directed the work, set the requirements (verification of every result, honest novelty assessment, no fabricated citations) and is responsible for its publication. All results were checked numerically as described in Section 6; nevertheless, readers should treat the novelty assessment, which is based on literature searches, with appropriate caution.

Code and data availability

All code used to derive and check the results, including the machine-readable a^6 and a^8 coefficients, is available at <https://github.com/JacobGoodchild/findformula2> (directory `code/`; scripts `univ_a2.py`, `univ_a4_fast.py`, `univ_polar.py`, `univ_check_light.py`, `univ_check_mass.py`, `univ_torque_check.py`, `kn_anyv_a2.py`, `kn_a4_check.py`, and the verification script `papers/paperA/verify_paperA.py`).

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